Smart Personalized Activity Routes for Kids (SPARK) - Progress Report

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Introduction

SPARK (Smart Personalized Activity Routes for Kids) is an interactive tool that helps parents and caregivers create activity schedules for children. Its aim is to empower users by providing a map-centric UI that makes it easy to organize schedules consisting of exciting and diverse activities that are also practical and support children's development.

Problem Definition

Today, parenting involves substantial cognitive labor, including planning family activities, coordinating schedules, and identifying opportunities for children's enrichment [1]. Research also shows that parents consider children's developmental and future-oriented needs when organizing their activities [2]. While there are existing platforms and tools that help separately address each of these challenges (e.g., scheduling apps and calendar websites, extensive research on beneficial activities for children at different ages), Team 61 has not seen a tool like SPARK that combines geographic data, age-specific developmental scoring, and multi-objective optimization to create personalized schedules for users in one application.

Literature Survey

Theme #1: Child Development and Caregiver Needs

Studies show the developmental benefits of child activities: active play has been associated with cognitive benefits [3], sports with psychological and social benefits [4], and child-led art with personal and social development [5]. Outdoor play affects executive functioning, emotional resilience, and gross motor skill development [6]. Organized extracurricular activities in general can improve academic scores, social-emotional well-being, more active lifestyles as adults and improved self-control [7]. Research has also shown that active play interactions can reduce familial stress. Conversely, these interactions may decline in quality and frequency when parents are stressed [8]. Chronic stress can negatively impact decision-making, leading to overly cautious or rigid practices [9]. These studies show the benefits of varied, deliberately planned activities for children but don't offer a transferrable framework for classifying activities and age bands.

Theme #2: Recommendation Systems, Data and Interface Design

Studies show that effective context-aware POI recommendation systems focus on displaying data relevant to users [10,11]. For SPARK, the relevant factors for activities are based on child-specific factors such as age, interests, developmental priorities, activity type, and practical constraints such as time and distance. High-quality geographic data is essential for SPARK. OpenStreetMap provides open-source POI data, but its completeness, accuracy, and timeliness vary [12,13]. For route optimization between POIs, the n-opt local search heuristic is a commonly used model that efficiently finds optimal paths for data sets as large or larger than SPARK requires [14]. The interface combines map-based activity discovery with scheduling, building on research showing the importance of family calendars [15] and task-focused map design [16].

Proposed Methods

Data extraction, transformation, and loading

POIs are extracted from an OpenStreetMap geopackage (gpkg) file. The geopandas module is used to read from the file and load it into the SQLite database. POI data is then analyzed and child activities' specific locations were extracted. Libraries are available as amenities while the rest of the POIs (parks, playgrounds, etc.) are available under a category named 'leisure'. Python is used to read data from the geopackage file and load into the SQLite database. The OSM ID of the POI is

used as the primary key to uniquely identify a POI. The file contains a multi-polygons layer to represent each location. Since latitudes and longitudes of the border of the polygon that represent a location are available, a representative point in the polygon, its latitude and longitude, is stored in the database. Separately, three mapping tables that map between POI categories and applicable activities were generated manually in Excel and then loaded into the same SQLite database. Scores across eight child-development factors per activity were calculated as well as age-dependent developmental priority weights for these factors. A final fifth table holds user ratings and comments for POI activites. This table is initially empty before first ratings are entered.

Regarding scores mapped to activities and age bands:
Each activity is evaluated against eight developmental factors: social-emotional development, cognitive-executive skills, language-communication, physical-motor development, exploration-creativity, foundational knowledge, independence-life skills, and health with scores from 0 to 10 (highest developmental benefit). Each age band is also scored 0 to 10 on the same factors to indicate importance of each factor for a given age band. These factors were selected based on research and guidance from CDC, Harvard Center on the Developing Child, California Infant-Toddler Learning and Head Start [17, 18]. Some activities were specifically mentioned in the sources and therefore received high scores in such areas. Any gaps were filled via collective judgement of the team. A final version of SPARK would ideally be reviewed by child development experts to ensure accuracy and validity of the model and assigned scores.

Data Integration and Candidate Generation
The SQLite database with its 5 tables is loaded in the central python notebook. The user's starting address is geocoded into latitude and longitude. The Haversine formula is then applied to calculate the geographic distance between the starting location and each POI. Only POIs within the user-defined search radius are retained. The resulting POIs are joined with the category-to-activity mapping and developmental scores to generate a set of candidate POI and activity combinations.

Multi-Objective Activity Selection
To generate a complementary agenda, SPARK employs a diversity-aware combinatorial optimization model. The objective function maximizes developmental coverage while penalizing geographic distance and redundancy between selected activities:

$$J(S) = D(S) - \delta L(S) - \mu R(S) + \gamma Q(S)$$

Where:

- Developmental coverage $D(S)$: Measures the combined coverage of the eight developmental factors, weighted according to the child's age. A diminishing-returns function is applied to prevent the agenda from over-investing in developmental factors that are already well covered.
- Location penalty $L(S)$: Penalizes agendas with POIs relatively far from the user's location.
- Redundancy penalty $R(S)$: Uses pairwise cosine similarity to identify similarities between activities and penalize combinations that offer largely overlapping developmental benefits.
- Quality/Rating bonus $Q(S)$: Favors agendas with higher ratings
- δ , μ , γ : Adjustable parameters controlling the relative importance of geographic proximity, developmental diversity and user ratings.

The optimization is performed using simulated annealing - a stochastic search algorithm. Starting from an initial feasible agenda using a set number of activities (specified by the user), the algorithm

iteratively replaces individual activities with alternative candidates and evaluates the resulting agenda using the objective function. Generally, the better solution is accepted, while worse solutions may also be accepted with a probability controlled by a gradually decreasing temperature parameter. This allows the algorithm to explore the solution space and reduce the likelihood of becoming trapped in a local optimum. The selection is subject to three constraints: the number of selected activities must match the user specified number of activities, no activity may appear more than once, and all selected POIs must fall within the specified search radius. The result is a prioritized selection of the user-specified number of geographically relevant and developmentally complementary activities, each associated with a specific POI.

Visualization and User Interaction

The SPARK UI is a two-panel dashboard built with Leaflet.js, D3.js, a local SQLite database, and is hosted through a lightweight Python server. Parents enter a starting address, child's age, travel radius, and desired number of activities. The left panel displays a recommended itinerary with a timeline and venue details, while the right panel shows an interactive OpenStreetMap with color-coded destinations and the suggested route. Users can further explore recommendations through the map or itinerary view, provide feedback on activities by upvoting or downvoting, and instantly see recommendations when trip settings change.

Innovations

Innovation 1 – Child Development Factor Model

Child-development research recognizes multiple, interconnected developmental domains and the importance of various activities for child development. To our awareness there is no framework to comprehensively translate these insights into a quantitative model for choosing everyday activities. SPARK bridges this gap by introducing an integrated, eight-dimensional development model with age-specific weighting of developmental priorities across a wide selection of activities. It transforms established developmental research and age-specific development guidance into a set of quantitative parameters that can guide and personalize activity selection as a child grows.

Innovation 2 – Comprehensively score a large set of daily activities across child development factors

Parents have access to an enormous variety of activities, but to our awareness there is no widely established, standardized methodology for comparing their potential contribution across multiple developmental domains. SPARK introduces a systematic, multidimensional scoring methodology that quantifies the potential developmental benefits of individual activities. SPARK creates a structured and comparable activity database, making it possible to identify activities that address specific developmental priorities and compare their broader developmental value.

Innovation 3 – Provide parents with a view on child activities in their neighborhood based on benefits for their child's development

Existing activity directories and recommendation platforms primarily help parents discover what is available nearby. Developmental guidance, meanwhile, is generally provided separately and requires caregivers to interpret and apply it themselves. SPARK introduces a child development-driven approach to local activity planning by integrating developmental priorities, activity-level scoring, geographic information and combinatorial optimization. Rather than simply recommending nearby activities, its algorithm identifies combinations of activities that collectively maximize developmental coverage and balance developmental value against distance and activity redundancy. The result is a personalized, location-aware activity agenda that translates developmental objectives into actionable experiences in a child's everyday environment.

Evaluation *(Under Construction)*

The below four key evaluation methods will be employed:

1. Manual model checks - Feasibility and sensibility of model-generated agendas will be checked . For each age band, distance (3 mile radius and 10 mile radius) and number of stops (3 stops and 6 stops) will be varied and it will be confirmed that there are no clear errors or suboptimal results like illogical routes or the same repeated activity at multiple locations. Any targeted actions will then be taken to improve methodology. The ratio of flawed agendas to total agendas tested will also serve as a KPI to monitor model performance on an ongoing basis.
2. User feedback - 10 test users will be surveyed via feedback survey form: 1) Do they consider SPARK generated agendas to be superior to their current child activity planning routine. 2) Do they save time planning activities using SPARK 3) Do the suggested activities make practical sense 4) Open feedback. This will indicate whether SPARK is able to provide better activity agendas, and if they save time planning activities with SPARK.
3. Programmatic agenda improvement – (implementation in progress) SPARK features the ability to rate activities at a given POI. The model incorporates ratings into the activity selection process and weighs activities-POI combinations more highly based on their rating. This will automatically improve recommended agendas over time as highly rated POI activities are more likely to be selected. It will also cause activities at POIs to be deprioritized that are technically beneficial to child development but unfavorable for other reasons, such as bad maintenance, insufficient space, overcrowding etc.
4. Evaluate scalability – The model thus far has been built using a smaller POI dataset focused on West Los Angeles. Larger datasets will be used to evaluate how well the model and visualization scale with more POI data. More extensive POI data will also allow us to test if the tool is able to create feasible agendas in more rural areas with a lower density of POIs.

Plan of Activities

Activity	Owner(s)	OLD Timeline	NEW Timeline	OLD Status	NEW Status
Literature review & project design	All	11/21-10/4	9/21-10/4	**Done**	**Done**
Identification of data sets	Max, Manoj, Shan	9/28-10/12	9/28-10/12	**Done**	**Done**
Data transformation and model selection	Max, Manoj, Shan	10/5-10/25	10/5-10/25	**In Progress**	**Done**
Develop initial models and UI	Shan, Manoj	10/25-11/15	10/25-11/15	**Not Started**	**In Progress**
Integrate models & UI	Shan, Manoj	11/9-11/22	10/28-11/22	**Not Started**	**In Progress**
Model evaluation and testing	Jay, Katherine	11/16-11/29	11/1-11/22	**Not Started**	**In Progress**
User surveys and feedback	Jay, Katherine	N/A (newly added)	11/22-11/29	**Not Started**	**Not Started**
Final report and poster development	All	11/23-12/3	11/23-12/3	**Not Started**	**Not Started**

All team members contributed a similar amount of effort and are expected to continue doing so.

References

[1] A. C. Weeks and L. Ruppanner, “A typology of US parents’ mental loads: Core and episodic cognitive labor,” *Journal of Marriage and Family*, 2025, doi: 10.1111/jomf.13057.

[2] M. Kremer-Sadlik, A. Fatigante, and C. Fasulo, “Making meaning of everyday practices: Parents’ attitudes toward children’s extracurricular activities in the United States and in Italy,” *Anthropology & Education Quarterly*, vol. 41, no. 1, pp. 35-54, 2010, doi: 10.1111/j.1548-1492.2010.01066.x.

[3] A. Minh, N. Muhajarine, M. Janus, M. Brownell, and M. Guhn, “A review of neighborhood effects and early child development: How, where, and for whom, do neighborhoods matter?,” *Health & Place*, vol. 46, pp. 155–174, Jul. 2017, doi: 10.1016/j.healthplace.2017.04.012.

[4] I. Jarvis, E. Gemmell, A. Cavanaugh, Z. Davis, M. Guhn, H. Sbihi, T. F. Oberlander, M. van den Bosch, and M. Brauer, “Neighborhood playability and early childhood development: A population-based birth cohort study,” *Environmental Research*, vol. 298, Art. no. 124124, Jun. 2026, doi: 10.1016/j.envres.2026.124124.

[5] M. Martin, A. Jelić, and T. D. O. Tvedebrink, “Children’s opportunities for play in the built environment: A scoping review,” *Children’s Geographies*, vol 21, no 6, pp. 1154–1170, 2023, doi: 10.1080/14733285.2023.2214505.

[6] M. Yogman, A. Garner, J. Hutchinson, K. Hirsh-Pasek, & R. M. Golinkoff, "The Power of Play: A Pediatric Role in Enhancing Development in Young Children,” *Pediatrics*, vol. 142, no. 3, Art. No. e20182058, 2018, https://doi.org/10.1542/peds.2018-2058

[7] D. L. Vandell and Z. Gülseven, “The Study of Early Child Care and Youth Development (SECCYD): Studying development from infancy to adulthood,” *Annu. Rev. Dev. Psychol.*, vol. 5, pp. 331–354, 2023, https://doi.org/10.1146/annurev-devpsych-120621-035345

[8] R. M. Eime, J. A. Young, J. T. Harvey, M.J. Charity, & W. R. Payne, “A systematic review of the psychological and social benefits of participation in sport for children and adolescents: informing development of a conceptual model of health through sport,” *The international journal of behavioral nutrition and physical activity*, vol. 10, Art no. 98, 2013 doi: 10.1186/1479-5868-10-98

[9] E. Theodotou, "Supporting personal and social development through child-led art projects in the early years settings,” E*arly Child Development and Care*, vol. 189, no. 11, pp. 1889–1900, 2019, doi: 10.1080/03004430.2017.1418739

[10] W. Shafqat and Y.-C. Byun, “A context-aware location recommendation system for tourists using hierarchical LSTM model,” *Sustainability*, vol. 12, no. 10, Art. no. 4107, May 2020, doi: 10.3390/su12104107.

[11] H. A. Rahmani, M. Aliannejadi, M. Baratchi, and F. Crestani, “A systematic analysis on the impact of contextual information on point-of-interest recommendation,” *ACM Trans. Inf. Syst., vol 40, no. 4, Art. no. 88, pp.1-35, 2022*, doi: 10.1145/3508478

[12] M. Haklay, “How good is volunteered geographical information? A comparative study of OpenStreetMap and Ordnance Survey datasets,” *Environment and Planning B: Planning and Design*, vol. 37, no. 4, pp. 682–703, Aug. 2010, doi: 10.1068/b35097.

[13] A. Psyllidis, S. Gao, Y. Hu, E.-K. Kim, G. McKenzie, R. Purves, M. Yuan, and C. Andris, “Points of Interest (POI): A commentary on the state of the art, challenges, and prospects for the future,” *Computational Urban Science*, vol. 2, Art. no. 20, Jun. 2022, doi: 10.1007/s43762-022-00047-w.

[14] S. Lin, "Computer solutions of the traveling salesman problem," in The Bell System Technical Journal, vol. 44, no. 10, pp. 2245-2269, Dec. 1965, doi: 10.1002/j.1538-7305.1965.tb04146.x.

[15] C. Neustaedter, A. J. B. Brush, and S. Greenberg, "The calendar is crucial: Coordination and awareness through the family calendar," *ACM Transactions on Computer-Human Interaction*, vol. 16, no. 1, Art. no. 6, Apr. 2009, doi: 10.1145/1502800.1502806.

[16] M. Bartling, A. C. Robinson, B. Resch, A. Eitzinger, and K. Atzmanstorfer, "The role of user context in the design of mobile map applications," *Cartography and Geographic Information Science*, vol. 48, no. 5, pp. 432–448, Jul. 2021, doi: 10.1080/15230406.2021.1933595.

[17] Office of Head Start, "Head Start Early Learning Outcomes Framework," HeadStart.gov, Dec. 23, 2024. [Online]. Available: https://headstart.gov/school-readiness/article/head-start-early-learning-outcomes-framework (accessed Oct. 21, 2026).

[18] California Department of Education, "California PTKLF," Early Education, Feb. 23, 2026. [Online]. Available: https://www.cde.ca.gov/sp/cd/re/psfoundations.asp (accessed Oct. 21, 2026).